

Project Executable Files

Date	04 July 2026
Team ID	SWTID-2026-1614
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example or similar)	No
6	Deployed application URL (if hosted)	Yes
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

```
hdi-predictor-ml-flask/
|-- Flask/
|   |-- templates/
|   |   |-- home.html           - Core input form template with searchable country dropdown
|   |   |-- index.html          - Legacy baseline form template
|   |   |-- indexnew.html       - High-fidelity glassmorphism results dashboard
|   |-- app.py                  - Main Flask server running routing and model inference logic
|   |-- update_dropdown.py      - Preprocessing helper script for country list validation
|-- Dataset/
|   |-- HDI.csv                 - Dataset containing historical indicators for 195 countries
|-- Training/
|   |-- HumDevIndex.ipynb       - Jupyter Notebook documenting data cleaning and training steps
|   |-- train_model.py          - Script to train and serialize the model to model.pkl
|-- .gitignore                  - Git exclusion settings
|-- requirements.txt            - List of dependencies (Flask, pandas, scikit-learn)
|-- README.md                   - Setup guide and project deployment instructions
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	https://hdi-predictor-ml-flask.onrender.com (or your custom deployment URL)
Login Credentials (Demo)	Not applicable – the application is openly accessible with no login required.
Platform / Hosting Provider	Render / PythonAnywhere
Repository Link	https://github.com/v-harishkumar/hdi-predictor-ml-flask
Demo Video Link	https://drive.google.com/file/d/your-video-id/view?usp=drivesdk (or your Drive link)

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

Pre-requisites:

Python 3.9+ and pip installed.

Step 1: Clone or download the project folder (hdi-predictor-ml-flask/).

Step 2: Install dependencies: `pip install -r requirements.txt`

Step 3: Ensure the pre-trained model file (model.pkl or similar) is present in the Flask/ folder.

Step 4: Run the server locally: `python Flask/app.py` (or: `flask --app Flask/app.py run`)

Step 5: Open your browser at `http://127.0.0.1:5000` (or the port shown in your terminal).

To deploy to Render instead of running locally:

Step 1: Create a free Web Service on Render linked to your GitHub repository.

Step 2: Set the runtime environment to Python and specify the build command

`pip install -r requirements.txt`

Step 3: Set the start command to `gunicorn Flask.app:app` (with gunicorn added to requirements.txt).

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Cold-start latency on Render: The first query can take ~30–50 seconds if the free container has fallen asleep due to inactivity.	Known limitation of free-tier hosting. Workaround: Set up a periodic keep-warm request or upgrade to an always-on instance.
2	Out-of-bounds regression values: Extreme user inputs can theoretically yield predicted HDI values slightly below 0 or above 1.	Workaround: Form input bounds constrain numbers to valid physical ranges (Life Expectancy 20-100, Schooling 0-25).
3	No authentication or user roles; the app is fully open access.	Identified as a planned future improvement for multi-user session storage. Acceptable for the current academic scope.